

Truss Design Report & Calculations and Performance Report

Team Gussy Bakas

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ENES102: Mechanics I

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****Materials Request Submission *Macallister Morris, Gavin Andrews***

****Contributions for the truss calculations are as follows***

1. Support Reactions - *Kieran Denyer*
2. Member Forces - *Kieran Denyer*
3. Internal Forces and Normal Stresses - *Kieran Denyer*
4. Buckling - *Arquinn Borden*
5. Shear Forces and Stresses - *Gavin Andrews and Macallister Morris*
6. Predicted Max Force - *Alan Qu*
7. Safety Factor - *Alan Qu*

*****Note: required for report, completed but not shown in document.***

Introduction to Truss Design

We will be constructing a 4-panel Pratt truss. We opted to choose a Pratt truss over other types of trusses to minimize members in compression and therefore prevent buckling. Buckling occurs easily for longer members, and the Pratt truss decreases the number of members in compression. With a centrally applied external load on the top of our structure, the longer inner diagonal members are placed in tension with this design, therefore minimizing buckling.

We chose four panels as the “sweet spot” for maximizing our structure’s strength/weight ratio within the given constraints. Using more panels (6+) would help by increasing the threshold for buckling due to shorter members; however, including more members means there is significantly less physical area for gusset plates to glue on to, as we only have 15% max for each member. Therefore, gusset plates would become more of an issue with more panels. Using less panels (2 panels) would violate the 6 member minimum requirement and increase the risk of buckling by increasing the length of our members. Therefore, four panels was our optimal design.

Material Property Values

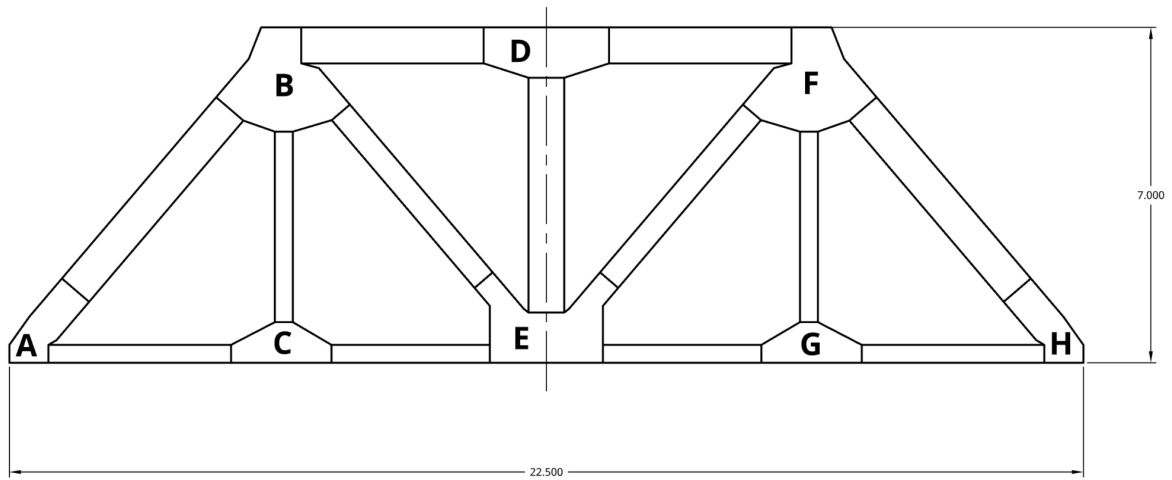


Fig. 1. Blueprint Diagram

Member(s)	Thickness (in)	Width (in)	Length (in)	Material	Cross-sectional Area (in ²)
AB/HF	0.75	0.75	7.07	Red Oak	0.5625
AC/HG	0.38	0.75	5.46	Pine	0.285
BD/FD	0.75	0.75	5.50	Red Oak	0.5625
BC/FG	0.38	0.75	5.67	Pine	0.285
BE/FE	0.50	0.75	6.60	Pine	0.375
CE/GE	0.38	0.75	4.73	Pine	0.285
DE	0.75	0.75	7.00	Red Oak	0.5625

<u>Material Properties</u>	Elastic Modulus (E) [psi]	Ultimate Tensile Strength (S_u) [ksi]
Red Oak	1.8×10^6	6.8
Pine	1.5×10^6	5.0
Birch Plywood	1.2×10^6	4.5

<u>Adhesive Properties</u>	Shear Stress (psi) on Oak	Shear Stress (psi) on Pine
Gorilla Glue	1.012×10^3	525

Technical Engineering Drawings/Blueprints

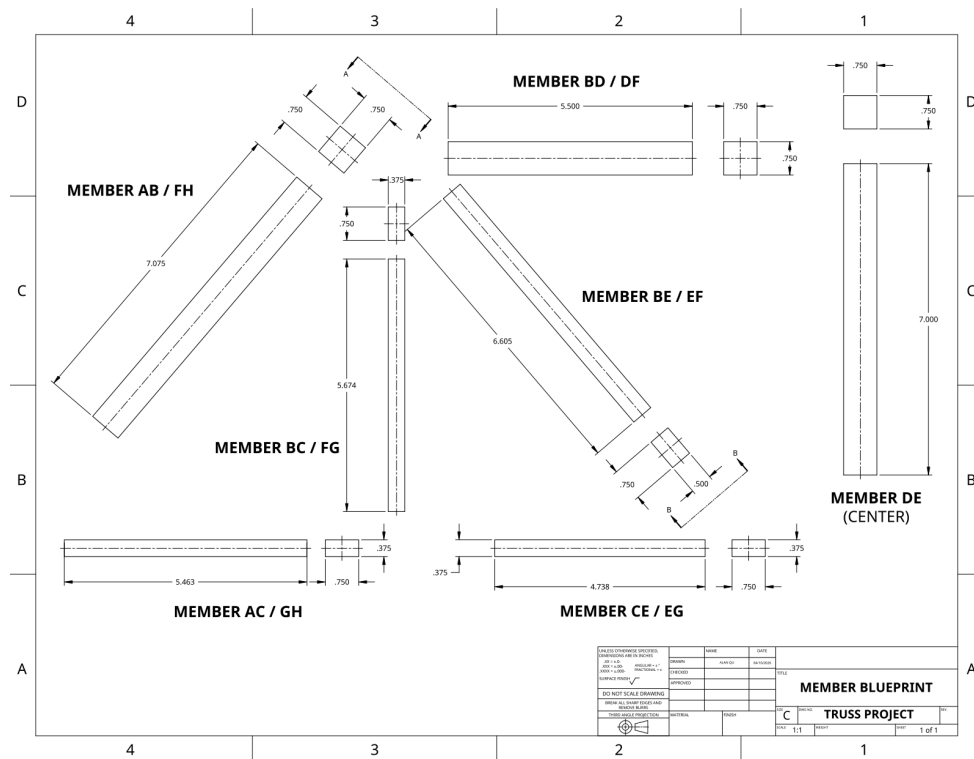
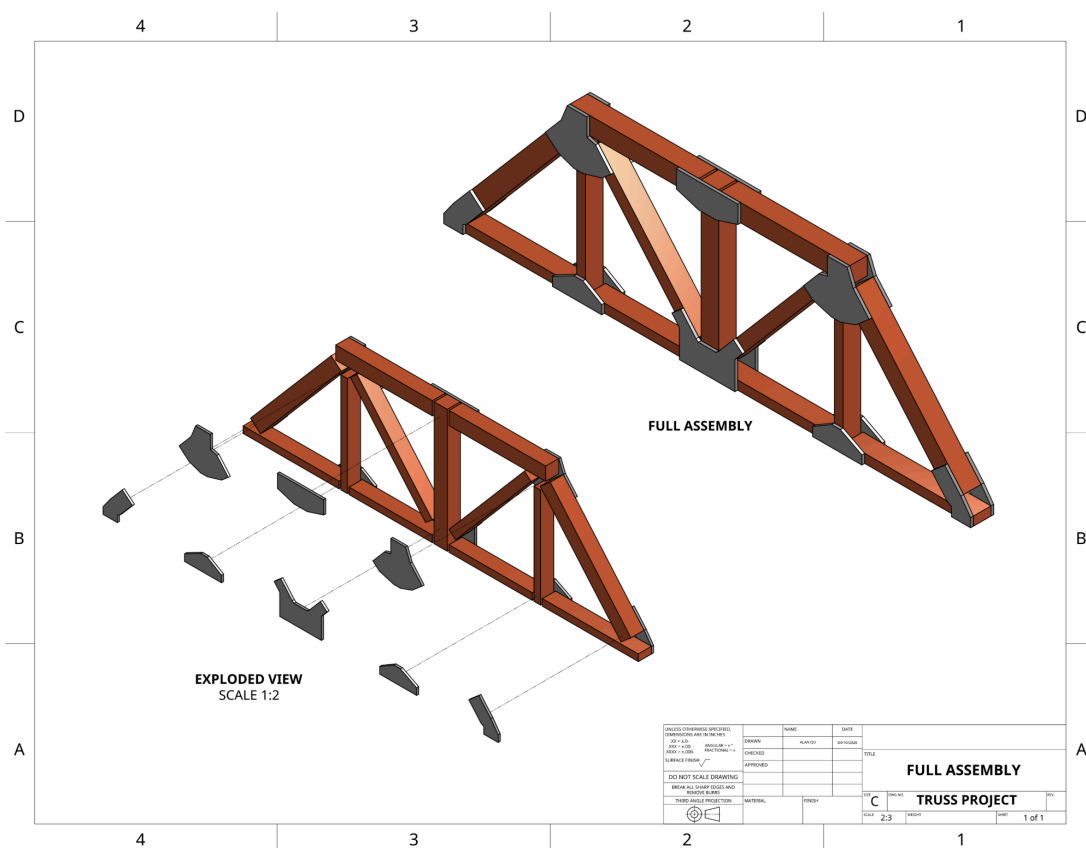


Fig. 2. Member Blueprint



Support Beam Justification and Characteristics

The drawings displayed are the specific parts utilized for each support member and each gusset plate. To reiterate, Members AB/FH, BD/DF, and DE utilize Red Oak. The reasoning for red oak stems from the fact that all the members at those locations are in compression. Red Oak is denser and provides better tensile strength and a higher elastic modulus (E) compared to pine. Properties of these materials are provided in the truss project instructions. Pine wood is utilized for the inner supports (AC/GH, BC/FG, CE/EG, BE/EF) as these members are in tension. Even though pine is generally “weaker” than red oak, its tensile strength is still more than enough to support the truss and the load without having to worry about buckling. Pine is also less dense compared to red oak, which provides weight-saving efficiencies.

The dimensions of the support beams are utilized to fit within the dimensional requirements of the project (22-23 inches long, 7 inches maximum height). The cross-sectional dimensions of the beams are not uniform, as stated in **the material property values**. The red oak members have the maximum cross-section of 0.75 inches by 0.75 inches to increase their critical buckling load. For most pine members, we chose the smallest cross-section possible (0.75 inches by 0.375/0.38 inches) to save weight. The cross-section of pine members BE/FE is slightly larger (0.75 inches by 0.5 inches) and is due to how the stress experienced by the two members is slightly higher than the other pine members; the thickness was increased to compensate for the increased load on these members specifically.

Gusset Plate Justification and Characteristics

The gusset plates, as specified, are made of birch plywood with Gorilla glue as the adhesive, per the project specification. Firstly, we placed gusset plates on both sides of our truss design. This halves the shear stress experienced by each joint by converting each joint from a

single to double shear joint ($\tau_{single} = \frac{V}{A}$; $\tau_{double} = \frac{V}{2A}$). The justification for each custom design, as provided in the Technical Engineering Drawings, specifies that the glue is distributed to exactly the maximum amount of member area coverage of 15%, whilst also distributing the applied force for each member to minimize shear stress (τ) experienced through increasing the surface area (A) on our double shear joints. ($\tau = \frac{V}{2A}$). The justification for the Gorilla Glue® adhesive for every gusset plate is that it is the strongest adhesive as compared to the alternatives. Gorilla Glue® offers a higher shear strength of 2500-3500 psi, making it the most optimal for the task required.

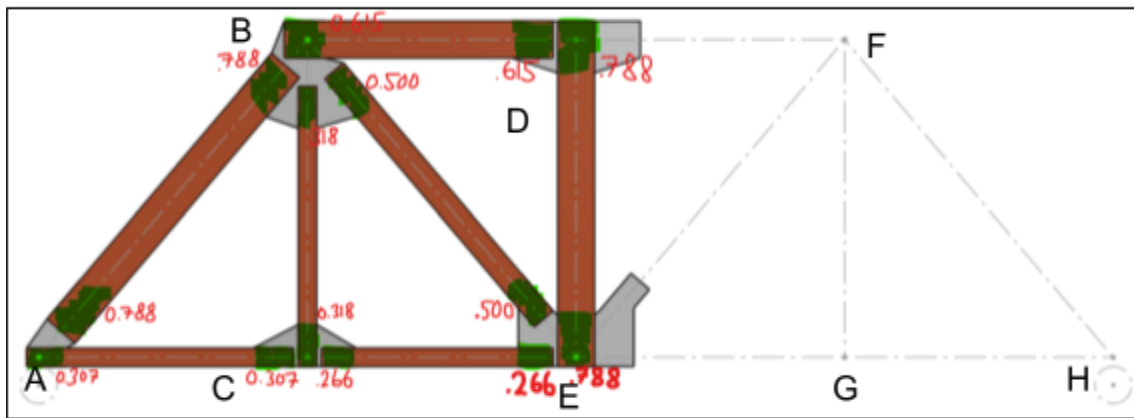


Fig. 5. Gusset Plate Areas

*Note: Areas in in² (in red) obtained by highlighting gusset-member overlaps in OnShape

Member(s)	Length (in)	Thickness (in)	Area	Max Area (Area x 0.15)	Gusset Plate Area
AB/HF	7.07	0.75	5.3025	0.795375	0.788
AC/HG	5.46	0.38	2.0748	0.31122	0.307
BD/FD	5.50	0.75	4.125	0.61875	0.615
BC/FG	5.67	0.38	2.1546	0.32319	0.318
BE/FE	6.60	0.5	3.3	0.5049	0.500
CE/GE	4.73	0.38	1.7974	0.26961	0.266
DE	7.00	0.75	5.25	0.7887	0.788

Truss Calculations and Performance Report:

See [Appendix](#) for link to LaTeX report.

Appendix:

- A) [Truss_Project_Final_LATEX_Calculations.pdf](#) - **LaTeX Calculations**